

A Deep Learning-based Cross-Dataset Exploration in LULC Classification Under Domain And Label Shift using Satellite Imagery

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Abstract—Land use and land cover (LULC) classification from remotely sensed satellite imagery is critical for urban planning, environmental monitoring, and disaster management. Deep convolutional neural networks (CNNs) now achieve strong performance on individual benchmark datasets, yet their ability to generalize across heterogeneous satellite datasets remains poorly understood. This work investigates cross-dataset generalization of deep CNNs under both domain shift (sensor characteristics, spatial resolution, atmospheric conditions) and label shift (differences in class taxonomies) using the EuroSAT and UC Merced Land Use datasets as a testbed. A set of ImageNet-pretrained architectures, including ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0, is evaluated in strictly in-domain, zero-shot cross-dataset, and few-shot fine-tuning regimes with varying numbers of labeled target samples per class. Experimental results show that models achieving above 90% accuracy in-domain can collapse to single-digit accuracy when transferred zero-shot to a different dataset, with only marginal gains under naive few-shot fine-tuning, even when up to 20 labeled samples per target class are available. The cross-dataset transfer is strongly asymmetric between EuroSAT and UC Merced, indicating that sensor resolution and class granularity interact with learned representations in a non-trivial manner. Across all regimes, architectural choices have a limited effect compared to the magnitude of domain and label shift. The findings highlight the inadequacy of simple supervised fine-tuning for remote sensing transfer and underscore the need for explicit domain adaptation and more robust representation learning approaches for operational LULC applications.

Index Terms—Cross-dataset generalization, Domain shift, Label shift, Few-shot learning, Remote sensing, Land use and land cover (LULC), CNN, Transfer learning.

I. INTRODUCTION

Land use and land cover (LULC) information of Earth's surfaces from satellite imagery is vital for urban planning, environmental monitoring, agricultural management, and disaster response [1], [2]. The rapid increase in the availability of high-resolution optical and multispectral data, along with recent advances in deep learning, has enabled substantial progress in automating LULC classification over the past

decade [3], [4]. Convolutional neural networks (CNNs) trained on carefully curated benchmark datasets can achieve very high in-domain accuracy, as shown in recent studies; they often surpass traditional machine-learning algorithms [3], [4].

Despite recent advances, most studies use training and test data that are drawn from the same underlying distribution. But in real-world scenarios, these models might have to operate across various sensors, resolutions, geographic regions, and landscapes that differ from those seen during training [4], [5]. When a model trained on one dataset is applied directly to another, differences in sensor characteristics, spatial resolution, atmospheric conditions, land-cover composition, and semantic taxonomies induce both domain shift and label shift, which can severely degrade performance. Although domain adaptation and domain generalization have been widely studied in computer vision and, increasingly, in remote sensing, the extent to which standard CNN baselines generalize across heterogeneous LULC benchmarks under realistic supervision regimes remains insufficiently explored.

Public benchmark datasets such as EuroSAT and the UC Merced Land Use dataset are central to evaluating LULC classification models and provide a natural testbed for studying cross-dataset generalization [6], [7]. EuroSAT consists of 10 LULC classes derived from Sentinel-2 multispectral imagery with 13 spectral bands and a ground sampling distance of 10–60 m, and has been extensively used to benchmark CNN architectures and transfer learning strategies for remote sensing [6]. In contrast, the UC Merced Land Use dataset contains 21 high-resolution aerial scene classes at 256×256 pixels and approximately 1-ft spatial resolution, originally introduced to evaluate bag-of-visual-words and spatial pyramid representations for land-use classification [7]. These two datasets differ markedly in sensor type, spatial scale, scene composition, and class granularity, which makes them well-suited for analyzing the impact of domain and label shift on deep LULC models.

Several recent studies have explored CNN-based approaches

for remote sensing scene classification and LULC mapping using EuroSAT and other related datasets, typically reporting strong in-domain performance [12]. Studies in remote sensing have proposed a variety of domain adaptation methods, including adversarial feature alignment, self-training, and style transfer, as well as few-shot and meta-learning frameworks to handle distribution shifts and limited target annotations [13], [14].

We focused on how the performance shifts under domain and label shift without applying any explicit domain adaptation methods. The zero-shot testing was done by loading models trained with one dataset and applying them against data from the other dataset without any target label, for both datasets. While in the few-shot adaptation, the trained backbone models were adapted to a small amount of labeled data per class.

This study explored the cross-dataset generalization of four ImageNet-pretrained CNN architectures, such as ResNet50, DenseNet-121, MobileNetV2, and EfficientNetB0, under domain and label shift between the EuroSAT and UC Merced Land Use datasets. Specifically, three evaluation regimes are considered: (i) strictly in-domain training and testing on each dataset; (ii) zero-shot cross-dataset transfer, where a model trained on one dataset is applied directly to the other without using any target labels; and (iii) few-shot fine-tuning, where only some labeled target samples per class are available. Upon learning the recent observations that CNNs can achieve high accuracy in-domain, this study explores the domain shift. We aim to explore the performance impact under a distribution shift. The study provides a focused empirical analysis of how in-domain performance translates, or fails to translate into cross-dataset robustness for LULC classification and discusses implications for the design of more reliable remote sensing systems [17].

II. RELATED WORK AND BACKGROUND

A. Remote Sensing LULC Classification

Classical LULC mapping was usually done by manual spectral indices, texture descriptors, and shallow classifiers such as support vector machines and random forests, often combined with pixel-wise or object-based image analysis [1], [2]. Deep CNNs have become the benchmark method for remote sensing scene classification and LULC mapping thanks to the development of large labeled datasets and GPU computing. As a handful of studies have shown, they outperform conventional machine learning algorithms. [4], [5]. EuroSAT is a benchmark dataset built from Sentinel-2 multispectral imagery. It has been widely used to evaluate CNN architectures and transfer learning strategies for optical LULC classification. [6] The UC Merced Land Use dataset and RESISC-45 provide high-resolution aerial and satellite scenes for benchmarking scene classification models [7]. These datasets have been used to train deep learning models for LULC classification. While

their in-domain accuracy is good, they only partially address the challenges of cross-dataset generalization.

B. Domain Shift and Label Shift

In practice, models trained on a single dataset often face substantial shifts when applied to new sensors, regions, or mapping products. Differences in spatial resolution, viewing geometry, atmospheric conditions, and acquisition protocols induce covariate or domain shift, while inconsistencies in class definitions and taxonomies lead to label shift [14]. Earlier works in computer vision showed that dataset bias can severely limit the transferability of features across benchmarks, and similar phenomena have been documented for remote sensing scene classification and LULC mapping [14], [15]. Recent cross-dataset studies indicate that even modern CNNs, when trained on one remote sensing dataset, can experience dramatic performance drops when tested on another without adaptation, underscoring the need to explicitly account for domain and label shift in evaluation protocols [16], [17].

C. Few-Shot and Meta-Learning in Remote Sensing

Few-shot learning and meta-learning algorithms have been adapted to remote sensing to address scenarios where only a small number of labeled examples are available for new classes, regions, or sensors. Metric-based and optimization-based meta-learning frameworks have been applied to remote sensing scene classification, demonstrating improved performance in low-shot regimes compared to naive fine-tuning of supervised baselines. More recently, generalized few-shot frameworks have been proposed for land cover mapping, combining semantic segmentation with prototype-based adaptation to discover and delineate novel classes from very limited supervision [16]. Despite these advances, most few-shot remote sensing studies focus on carefully designed episodic benchmarks and do not explicitly analyze how such methods behave under strong domain and label shift between widely used LULC datasets.

D. Summary and Positioning

Overall, prior work has established that deep CNNs can achieve high in-domain accuracy on benchmark LULC datasets and that domain adaptation and few-shot learning methods can partially mitigate data scarcity and distribution shift. However, there remains a gap between sophisticated adaptation frameworks and simple but practically relevant scenarios in which a model trained on one standard dataset must be deployed on another with no or very few target labels. This study explores this gap by focusing on cross-dataset generalization between EuroSAT and UC Merced using standard CNN architectures, explicitly quantifying in-domain, zero-shot, and few-shot performance without employing specialized domain adaptation techniques.

III. METHODOLOGY

A. Problem Formulation

Given a source dataset $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and a target dataset $\mathcal{D}_t = \{(x_j^t, y_j^t)\}_{j=1}^{N_t}$ with different input and label distributions, this study aims to evaluate cross-dataset generalization of CNN classifiers under three regimes: in-domain training, zero-shot transfer, and few-shot fine-tuning. In the in-domain setting, a model is trained and evaluated on the same dataset to establish an upper bound on achievable performance under matched conditions. In the zero-shot setting, a model trained on $\mathcal{D}_s$ is directly applied to $\mathcal{D}_t$ without using any target labels, revealing the impact of domain and label shift. In the few-shot setting, only a small labeled subset of $\mathcal{D}_t$ is available for fine-tuning, which approximates realistic scenarios where limited annotation effort can be invested in a new region or sensor [6], [7], [12], [13].

B. Datasets

1) *EuroSAT*: EuroSAT is a Sentinel-2 based LULC dataset with 10 classes and 13 spectral bands at ~ 10 m ground sampling distance, covering diverse European regions and land cover types [6]. Following common practice, RGB bands are extracted to construct three-channel inputs, and images are resized to a fixed spatial resolution for CNN processing. The dataset has been widely used to benchmark deep architectures and transfer learning strategies for optical LULC classification [9], [10].



Fig. 1: Representative EuroSAT RGB patches illustrating different land use and land cover classes.

2) *UC Merced Land Use*: The UC Merced Land Use dataset contains 21 land use classes with high spatial resolution aerial imagery at 256×256 pixels and approximately 1 ft spatial resolution, originally collected from the USGS National Map Urban Area Imagery collection [7]. It's a 21-class dataset which somewhat aligns with EuroSAT categories while retaining a diverse set of scenes, including residential,

agricultural, and natural environments. Images are converted to three-channel RGB and resized to match the input size used for EuroSAT.

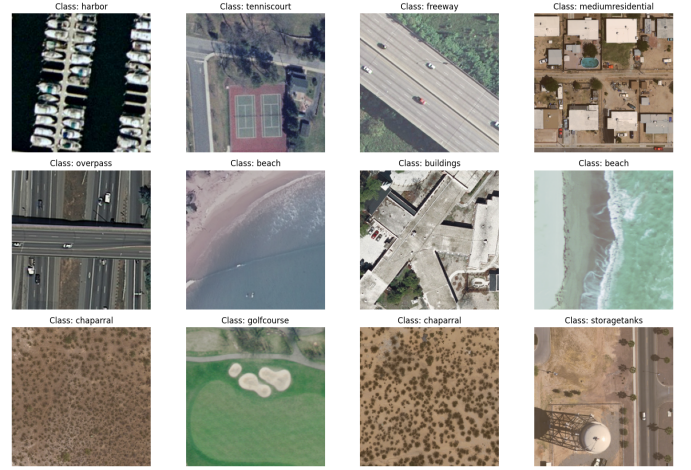


Fig. 2: Representative UC Merced Land Use RGB patches illustrating selected scene classes.

3) Preprocessing:

a) *Image Resizing and Color Channels*: To enable cross-dataset experiments, all images were resized to a common spatial resolution (128×128) in pixels and converted to three-channel RGB inputs. For EuroSAT, RGB bands were selected from the original Sentinel-2 multispectral imagery, and the resulting patches were resized to the network input size used for all models. For UC Merced, high-resolution aerial images were also resized to the same spatial dimensions as EuroSAT patches.

b) *Dataset Splitting*: Each dataset was split into training, validation, and test samples using a 60/20/20 ratio on a per-class basis. We used a fixed random seed to ensure reproducibility. The validation set was used for monitoring training, and the test set was used for final evaluation.

c) *Normalization and Data Augmentation*: All images were normalized to the $[0, 1]$ range by dividing pixel values by 255, ensuring a consistent input scale across both datasets and compatibility with the pretrained CNN backbones. During training, standard data augmentation was applied, including random horizontal and vertical flips, small rotations, and slight translations, to improve robustness and reduce overfitting. No augmentation was applied at test time, and the same preprocessing pipeline is used for both datasets to ensure comparability across experiments.

C. Model Architectures

We picked four ImageNet-pretrained CNNs: ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0, which provide a spectrum of depth, width, and parameter efficiency while remaining widely used in remote sensing applications [8], [10]. Each backbone was modified before its original

classification head, and a new fully connected layer with softmax activation was added to match the number of classes for the dataset under consideration. For the in-domain training, we applied fine-tuning after training each model by unfreezing initial layers.

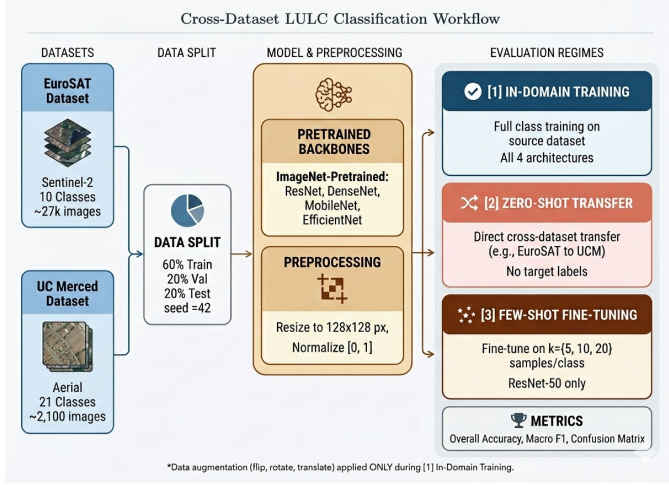


Fig. 3: Schematic overview of the Architecture

D. Training Protocols

1) *In-Domain Training*: For each dataset, models are trained and evaluated using train/validation/test splits to establish in-domain baselines. Training uses the cross-entropy loss, the Adam optimizer with an initial learning rate selected via validation, and early stopping based on validation loss to prevent overfitting. Standard data augmentation, including random flips, rotations, and small translations were applied to improve robustness. We trained each models with 20 epochs while applying early stopping to minimize computational constraints. Callbacks and Model Checkpoint using Keras were applied to log the best weights and save the best models.

2) *Zero-Shot Cross-Dataset Transfer*: In the zero-shot setting, a model is trained on the source dataset $\mathcal{D}_s$ using in-domain pre-trained models and then directly evaluated on the target test set from $\mathcal{D}_t$ without any access to target labels during inference or hyperparameter selection. The source model's classification head retains the number of output neurons from the source domain (10 for EuroSAT-trained models; 21 for UC Merced-trained models). Predictions on $\mathcal{D}_t$ are interpreted by taking the argmax over the source model's fixed output logits. Performance is summarized using accuracy and macro-averaged F1-scores. This configuration isolates the impact of domain and label shift on the learned representations [14].

3) *Few-Shot Fine-Tuning*: For each transfer direction, a small number of labeled subset of the target dataset is sampled with $k \in \{5, 10, 20\}$ examples per class. Few-shot fine-tuning experiments are conducted using ResNet50, the best-performing in-domain architecture, as the source backbone.

The fine-tuned model is then evaluated on the held-out target test set. [16].

E. Evaluation Metrics

We report overall classification accuracy and macro-averaged F1-score for all regimes. Overall accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}, \quad (1)$$

where TP denotes true positives, TN true negatives, FP false positives, and FN false negatives. Precision (P) and recall (R) are further used to assess the effectiveness of the classifiers:

$$\text{Precision}(P) = \frac{TP}{TP + FP}, \quad (2)$$

$$\text{Recall}(R) = \frac{TP}{TP + FN}. \quad (3)$$

The F1-score combines precision and recall into a single harmonic-mean measure:

$$\text{F1-score} = \frac{2 \times P \times R}{P + R}. \quad (4)$$

In addition to these global metrics, macro-averaged F1-score is computed by averaging the class-wise F1-scores over all C classes:

$$\text{F1}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c. \quad (5)$$

In addition, confusion matrices and per-class accuracies are computed to analyze error patterns and identify systematically confused categories in both in-domain and cross-dataset experiments.

F. Implementation Details

All experiments were implemented in TensorFlow/Keras using an Nvidia Tesla T4 GPU provided by Google Colab. Training hyperparameters such as learning rate, batch size, and number of epochs are selected based on validation performance on the source dataset and kept fixed across transfer directions to ensure comparability. For few-shot experiments, fine-tuning is limited to a small number of epochs to prevent overfitting to the limited target data. Random seeds are fixed for data shuffling and weight initialization to improve reproducibility.

IV. RESULTS AND ANALYSIS

A. In-Domain Baselines

As expected, the in-domain results were excellent. It verifies the findings of previous studies, proving that transfer learning achieves high accuracy across all classes. This study even surpasses the findings of a recent study showing a small accuracy gain when we're conducting in-domain testing [4], [5].

Table I shows in-domain performance for each architecture and dataset, using the 60/20/20 train/validation/test splits described in Section III. Across both EuroSAT and UC Merced, all four CNN backbones achieve high accuracy and macro-averaged F1-scores, confirming that standard ImageNet pretrained architectures are capable of fitting each dataset when training and test distributions are matched. These results serve as reference points for interpreting subsequent cross-dataset experiments, where performance gaps can be attributed primarily to domain and label shift rather than model capacity.

TABLE I: In-domain classification performance on EuroSAT and UC Merced for different CNN architectures.

Dataset	Model	Accuracy	Macro F1
EuroSAT	ResNet50	0.9850	0.9844
EuroSAT	DenseNet121	0.9817	0.9812
EuroSAT	MobileNetV2	0.8550	0.8535
EuroSAT	EfficientNetB0	0.9756	0.9747
UC Merced	ResNet50	0.9476	0.9470
UC Merced	DenseNet121	0.9452	0.9446
UC Merced	MobileNetV2	0.9167	0.9188
UC Merced	EfficientNetB0	0.9405	0.9364

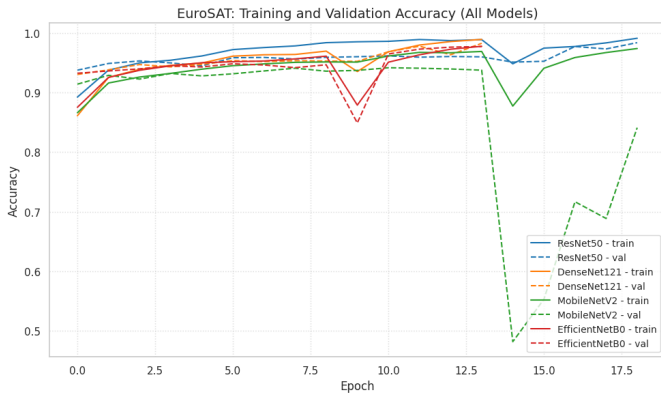


Fig. 4: EuroSAT training and validation accuracy per epoch for all CNN architectures.

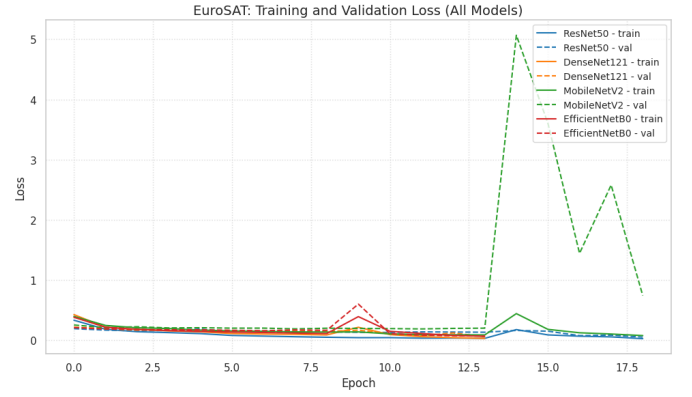


Fig. 5: EuroSAT training and validation loss per epoch for all CNN architectures.

The accuracy and validation curves show how the models behaved. ResNet50 and DenseNet121 are the most consistent, with stable convergence and no major sign of overfitting. MobileNetV2 on the other hand saw major shift in accuracy/loss curve during the initial phases of fine-tuning which impacted its overall accuracy. Where EfficientNetB0 was somewhere in the middle while maintaining good accuracy and showed stable gains/loss.

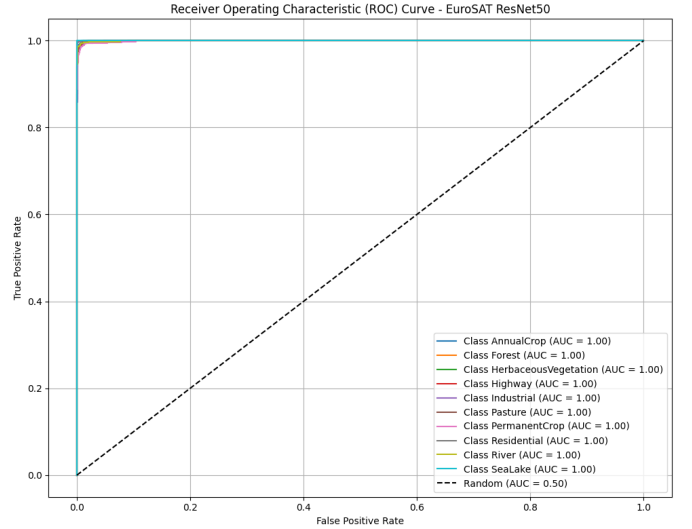


Fig. 6: EuroSAT macro-averaged ROC curve for the best in-domain model (ResNet50).

The Receiver Operating Characteristic (ROC) analysis for the best performing model ResNet50 demonstrates perfect discriminative performance across all classes in the EuroSAT dataset. Each class achieving an Area Under the Curve (AUC) of 1.00. This is consistent with the stable convergence and high accuracy observed during training. It confirms that the model learned highly separable features for each class without significant confusion.

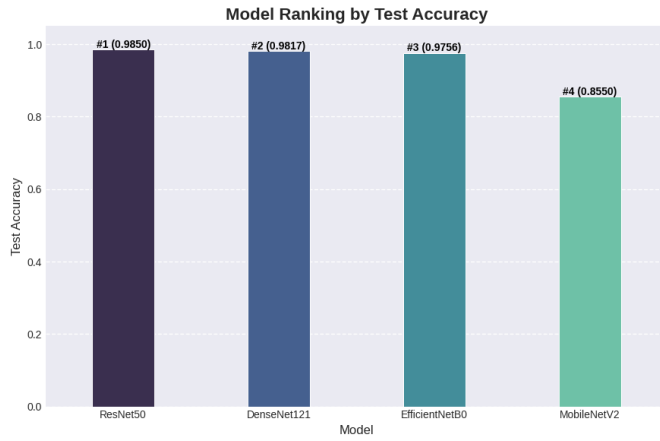


Fig. 7: EuroSAT in-domain performance comparison of all CNN architectures

The final comparative performance of the four benchmarked architectures is summarized in Figure 8, which ranks the models by their test set accuracy. This ranking provides a definitive, practical hierarchy of model effectiveness for the EuroSAT classification task.

The study on the UC Merced dataset result show different convergence patterns compared to EuroSAT. While most models learn effectively, their stability, final performance, and susceptibility to overfitting differ across models.

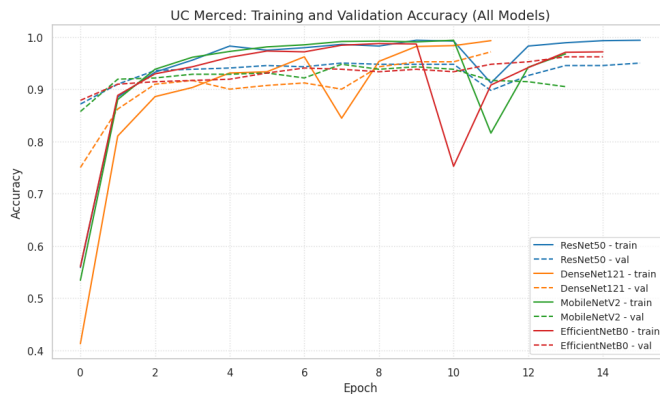


Fig. 8: UC Merced training and validation accuracy per epoch for all CNN architectures.

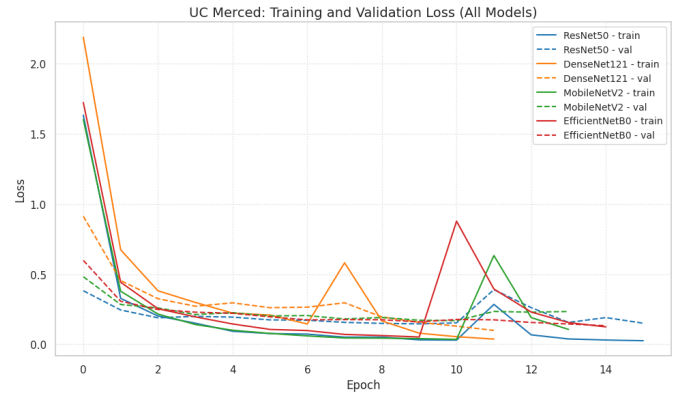


Fig. 9: UC Merced training and validation loss per epoch for all CNN architectures.

The results across both the EuroSAT and UC Merced datasets consistently identified ResNet50 as the top-performing model. It scored the highest test accuracy here as well with stable convergence and minimal overfitting. DenseNet121 reliably placed second, demonstrating strong performance. EfficientNetB0 showed competitive accuracy but showed some instability. MobileNetV2 scored the lowest accuracy across all models again.

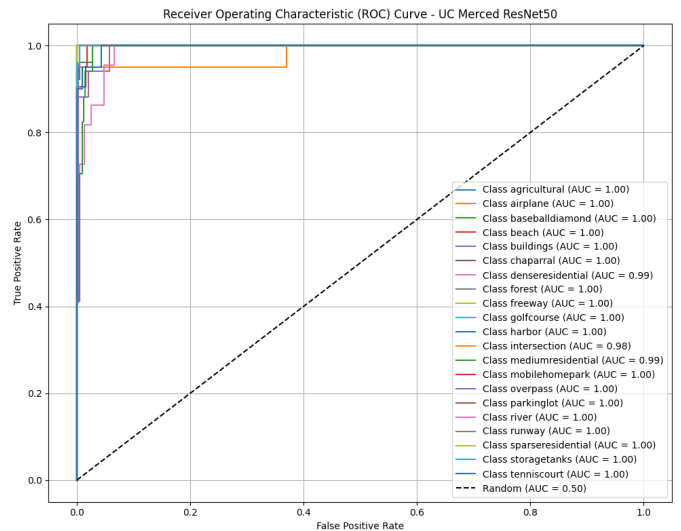


Fig. 10: UC Merced macro-averaged ROC curve for the best in-domain model (ResNet50).

The ROC curve confirms that ResNet50's high accuracy on the UC Merced dataset. While most classes are perfectly separable, the slight dip in AUC for intersection, dense residential, and medium residential highlights the model's minor difficulty with visually similar urban infrastructure.

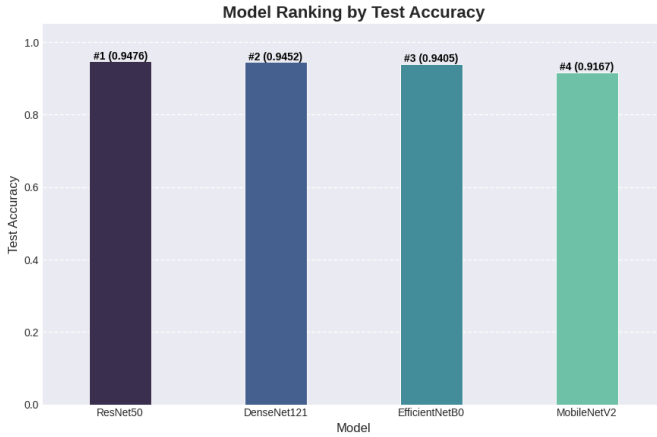


Fig. 11: UC Merced in-domain performance comparison of all CNN architectures

The final comparative performance of the four benchmarked architectures on the UC Merced dataset is summarized in Figure 13, which ranks the models by their test set accuracy. This ranking establishes a clear, practical hierarchy of model effectiveness for the land-use classification task

B. Zero-Shot Cross-Dataset Transfer

Table II summarizes zero-shot transfer from EuroSAT to UC Merced and from UC Merced to EuroSAT. Despite strong in-domain baselines, all models collapse to single-digit accuracy and near-zero macro F1-scores when evaluated directly on the other dataset without any adaptation, confirming the impact of domain and label shift.

TABLE II: Zero-shot cross-dataset performance. Accuracy and macro F1 are reported on the target test set without using any target labels during training.

Direction	Model	Accuracy	Macro F1
EuroSAT $\rightarrow$ UCM	ResNet50	0.038	0.003
EuroSAT $\rightarrow$ UCM	DenseNet121	0.074	0.032
EuroSAT $\rightarrow$ UCM	MobileNetV2	0.060	0.042
EuroSAT $\rightarrow$ UCM	EfficientNetB0	0.079	0.023
UCM $\rightarrow$ EuroSAT	ResNet50	0.093	0.017
UCM $\rightarrow$ EuroSAT	DenseNet121	0.070	0.026
UCM $\rightarrow$ EuroSAT	MobileNetV2	0.095	0.021
UCM $\rightarrow$ EuroSAT	EfficientNetB0	0.113	0.020

Overall, the zero-shot results show that high in-domain accuracy does not translate into cross-dataset generalization: all four CNN backbones collapse to accuracies below 12% and macro F1 scores near zero when evaluated directly on the other dataset. This failure is asymmetric, with EuroSAT $\rightarrow$ UC Merced transfer consistently weaker than UC Merced $\rightarrow$ EuroSAT, suggesting that models pretrained on the more heterogeneous UC Merced scenes retain slightly more reusable structure for EuroSAT than the other way around.

C. Few-Shot Fine-Tuning

Table III shows few-shot fine-tuning results with $k \in \{5, 10, 20\}$ labeled target samples per class for the ResNet50 backbone. Even with access to a small number of target labels, improvements over the zero-shot baselines remain modest, particularly for the EuroSAT $\rightarrow$ UC Merced direction.

TABLE III: Few-shot fin-tuning performance of ResNet50 for $k \in \{0, 5, 10, 20\}$ labeled target samples per class. Accuracy and macro F1 are reported on the target test set.

Source $\rightarrow$ Target	Model	(k)	Accuracy	Macro-F1
EuroSAT $\rightarrow$ UCM	ResNet50	0	0.0381	0.0035
		5	0.0476	0.0043
		10	0.0476	0.0043
		20	0.0476	0.0043
UCM $\rightarrow$ EuroSAT	ResNet50	0	0.0933	0.0171
		5	0.1083	0.0195
		10	0.1167	0.0209
		20	0.1167	0.0209

The few-shot experiments indicate that limited access to target labels only partially mitigates this domain gap. For EuroSAT $\rightarrow$ UC Merced, increasing k from 0 to 20 yields less than one percentage point improvement in accuracy and almost unchanged macro F1, while UCM $\rightarrow$ EuroSAT sees somewhat larger gains but still remains far below in-domain performance. These trends imply that, under strong domain and label shift, a small number of labeled target examples is insufficient to recover the rich decision boundaries learned from fully supervised training on the target dataset.

V. DISCUSSION AND LIMITATIONS

A. Interpreting Zero-Shot Failures

Zero-shot cross-dataset performance collapses due to severe shifts in spectral information, spatial resolution, and class semantics between EuroSAT and UC Merced. Even strong in-domain baselines with accuracies above 95% degrade to single-digit cross-dataset accuracy and near-zero macro F1 when evaluated without any target labels, confirming that source features do not linearly separate target classes. The particularly low scores for EuroSAT $\rightarrow$ UCM suggest that models overfit to the characteristic textures, acquisition geometry, and class taxonomy of their training dataset, and that naïve reuse of supervised features across sensors and label spaces is insufficient for reliable land cover mapping.

B. Few-Shot Gains and Saturation

Few-shot fine-tuning provides only marginal improvements and quickly saturates, suggesting that the underlying representation remains misaligned with the target domain. For ResNet50, adding 5-20 labeled samples per class yields gains of at most 1-2 percentage points in accuracy and very small macro-F1 improvements, with the EuroSAT $\rightarrow$ UCM direction saturating already at $k = 5$. These results indicate that a small

number of labeled target examples can locally adjust decision boundaries but cannot restructure a representation that was learned under a different spectral, spatial, and semantic regime.

C. Limitations

This study is limited to two datasets, a small set of architectures, and a straightforward fine-tuning procedure. The experiments do not consider more recent vision backbones, explicit domain adaptation methods, self-supervised pretraining, or multi-dataset training, all of which could substantially alter the observed transfer patterns.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This work systematically evaluated cross-dataset generalization of CNNs for remote sensing LULC classification using EuroSAT and UC Merced Datasets. In-domain experiments confirmed that standard backbones such as ResNet50, DenseNet121, MobileNetV2, and EfficientNetB0 can achieve accuracies above 90% and macro-F1 scores in the same ballpark when training and test data are from the same distribution. However, these strong baselines collapse under zero-shot cross-dataset transfer, where accuracies drop to the single-digit range and macro-F1 scores approach zero, demonstrating that high in-domain performance does not guarantee robustness under realistic domain and label shift. Few-shot fine-tuning experiments with ResNet50 showed that providing 5-20 labeled target samples per class yields only modest improvements. This proves that LULC classification under domain and label shift without strict adaptation techniques poses a challenge in automating remote sensing tasks.

B. Future Work

Future research should explore more principled domain adaptation frameworks that explicitly align source and target distributions while preserving class structure. Vision Transformer and Data Fusion can greatly enhance remote sensing tasks that future studies can explore.

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